Karl Friston: Active Inference and Deep Temporal Models

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thank you very much for not introducing me thank you great pleasure to be here when my favorite places there's some of my favorite people I mean you're all my favorite people is it someone I'm not my favor you're not here so there are so many points of contact that I can't keep more in my head I'm going to go back to basics a bit and just review active inference and the free-energy principle in a way that speaks to a lot of the questions in particular last few questions we've just had about how we can migrate insights about perception and predictive processing to I think the deeper question of active inference and querying our world giving as an agency answering questions about how do we choose the sparse data in an optimal way to build oh I see that's the one yeah can you hear me now right what was I saying I was probably said all the importance action that's right so we're going to we're going to go from predictive processing to active inference and on route I think touch upon a lot of the key issues that we've just heard about from a formal perspective from a maths from a physicist perspective everything that I'm going to talk about it's exactly consistent with what I imagine from a philosophical perspective you were saying in your talk it also speaks very closely to some of the more formal issues about the neuronal code and what implications that may have for the sorts of contents the things that we can represent and happily I'm also even going to do dogs and cats and variational what we encode is as well so that's all good I should confess that I've got enough material for about an hour and a half so what so why I'm going to ask my chair to do five minutes before my allotted 30 minutes if you can tell me and then what I'll do is I'll just literally race through the slides to show you what you know I'm serious about this because all the interesting stuff is at the beginning but just you know have this been a more informal workshop what you would have seen so that we get plenty of time for discussion so seriously just I'll just stop when we get to get to that point so the interesting bit at the beginning active influence and self evidencing and I'm using actually influence deliberately here because I repeat it's gonna be about how we as a self go and solicit gana evidence for our own internal models our generative models of the world I mean going to call that active infants we're not going to use reinforcement learning we're going to use exactly the same objective function that the variational autoencoders use variational free-energy and we're going to put into it or we're going to unpack that objective function in a way that allows us to answer questions about how do you put prior preferences in how do you do the epistemic properly I think underneath all this though it's all about me interrogating my world so there's a sense in which that's the interesting stuff and in light of that we're going to talk we've rehearsed briefly don't in bottles but much of the hard work has already been done thank you very much even down to the hidden Markov model so you can skip through those the stuff that I won't show that I will skip through deliberately is really fun neurobiologists that you know you've got these ideas in play this formalism this math how would it materially inform your experiments what you'd look for with electric f log Rafi or psycho physics but we weren't really focusing on that if I have time I'll just show you what what I would have shown you in terms of state of the art in terms of using these sorts of schemes by simulating reading so composing deep hierarchical hidden Markov models work out where to look next to gather the rent the best sort of evidence for what you believe is a current narrative in play that's generating your sensory impressions right so very quickly you've all seen this sort of thing before the idea being that we're going to predict in everything on the notion of a generative model your perception is basically what is inside your head and sensations are just in the game of providing evidence for or against those beliefs I like this example from I think this is the 16th century famous oil painter who would paint these still lives that when you invert the picture they transpire to be faces and the point that I would normally make here now if I have more time is of course that percept is inside your head that's all we're actually starting where you ended with your pretty faces these ideas you are more than familiar with some of the key architects of things like deep learning also figure in terms of the other sort of brief history of the brain as a statistical organ generator of hypotheses or fantasies nicely summarized here by Helmholtz and the notion that objects are always imagined as being present in the field of vision as would have to be there in order to produce the same impression on the nervous system you've probably rehearsed many times the psychological gloss on that in terms of deception as hypothesis testing that getting into machine learning via people like Peter Dyer and Geoffrey Hinton lots of lots of people boring from Bayes and Bayesian probability theory and statistical physics to come up with notions of the Helmholtz machine which were parallel so this is the old school I would imagine before it got overshadowed by deep learning but very much these ideas have dealt in parallel and if I understand your talk properly now the generative models are coming back again and starting to address you know that the new technology or the old equations with the new technology are now being turned to focus on some of the questions that were raised in machine learning by Geoffrey Hinton in the sort of late 80s early 90s so this notion as objects being in the field of vision others would have to be there to nervous system very reminiscent of sensory impressions of shadows on Plato's cave and our job the brains job is then just to try and come up with a good explanation for those sensory patterns there's sensory sensory shadows and I'm using the next slide just to introduce now the formalism and I you know I don't apologize for this if you don't like maths take these things as iconic if you do like maths I think this is really useful I think it grounds questions that we were dancing around with wondrous presentation so if you have a formal understanding there's a simplicity that you can appeal to in in providing various visit ancestors and the questions that were being posed to the presenters earlier on so here the dogs and cats again the idea here is that were given this sensory shadow say that I was support my retina was supplied with these this data and I have to come up now with an explanation for what caused these sensory inputs and I'm going to frame that coming up with that process of Bayesian inference model selection which I haven't heard but I think it's a very relevant concept in terms of the unitary aspect of the contents of consciousness we have to select the best hypothesis on the basis of the evidence for that hypothesis inherent in the data in variational approximate Bayesian inference that's a variational free-energy hence a variational autoencoder so I just want to unpack what that free at that variation free energy is and I'm going to use this notation so Q is a belief is a probability distribution over discrete or continuous States and I use discrete States in let's talk at a particular time and all I'm going to do is maximize this free energy functional it's negative free energy in physics so what is it well it's just the evidence for my particular hypothesis - a term called the divergence which I can rewrite the divergence between the true posterior belief the best beliefs about the labels or the causes of my observed outcomes or my data in relation to this belief structure that I'm trying to optimize crucially I'm going to rewrite that and in terms of the difference between accuracy and complexity and the reason I'm making a big fuss about this is that it's useful to concentrate on this this term I think everybody I will be comfortable with if you like predictive coding or predictive processing Casper's predictive coding this is essentially just the seller squared prediction error so this is the prediction error that we're trying to minimize this I think is very the more interesting term it's the complexity is how much we have to move our beliefs away from the prior in order to offer an accurate explanation for the data and it effectively schools the degrees of freedom the dimensions on a manifold the number of elements in a hidden state required to compress all that good information down so that I can now generate a fantasy or a prediction that is as accurate as is possible so those variational auto-encoders they have something which is you'll see right through the history of things like machine loans to sticks they have a bottleneck architecture you're forcing the data through a bottleneck with low complexity simple low degree degrees of freedom representation minimum complexity just to keep you Evan Schmitt Hooper happy and then expanding it again to actually reproduce the thing that you're trying to predict that complexities a repeat can play a very important role when we move to the next level of argument which is how we actually sample the next bit of data in order to confirm or disconfirm our beliefs but just an illustrative example cats and dogs so what calls this was this a canine dog or a wolf or was it a little pussycat or and a bird and worry and even what do you want cat you see this before us sorry he's gonna go with the dog thank you well-behaved member of the audience sir right so we have we have a number of hypotheses in mind we're going to select this one as providing the most accurate explanation for these sensory impressions that próximas is the accuracy in this instance now after about the complexity and that's basically one can rewrite this in terms of prediction errors and you can get a nice predictive coding scheme out of this you can also unfold this know let's not go there to make it to make a variation or to encoder but let's not go over so that's the basic idea so we're focusing in this instance on object up tamai zhing and we can choose many different schemes to do that a free energy functional of the data and our beliefs about those data so it's all about how our brains are entertained engaged with the scent the sensory sampling of the world and of course it was a cat you're absolutely right but he did seem that he'd seen the the slide before so the point being of course you will never know what's actually out there your job is just to come up with the most plausible hypothesis so this was the illusion of a dog it wasn't actually a dog but it's still the Bayes optimal explanation even though it was completely wrong right so here's the his the the the key move then getting from classification based upon genitive models by via optimizing a variation free energy functional to active inference and self evidencing so there here is there are two ways that we can optimize this description of the Leo the goodness of our inference we can either change these beliefs encoded by our brain and a way yet to be determined to minimize this divergence to make our beliefs much much more like the posterior distribution much like what to make our beliefs consistent with their observations or we can actually change the data that are sampled to optimize the evidence and that's the key thing the what this one function has in it everything necessary to prescribe both the perception and the sampling and the behavior of me practically what that boils down to is this changing beliefs conditioned upon a policy to maximize this free energy functional of of a policy and changing sensations to maximize evidence the only way we can change this through action is by changing the outcomes and the observations and crucially what we're going to do is posit some beliefs over policies that are or let me first of all make a key observation here what we in order to count for action we now have to supplement these beliefs about states of affairs in the world with beliefs about what I'm going to do and that brings to the table two of that well yes two things but effectively the same thing these are these are the reactive dispositions that Andy was talking about these are how would I react how would I act how would I act and what are my P dispositions - they're acting so they're going to be the prior beliefs about how I would behave in this instance so that PI that policy is gonna be a sequence of actions bringing dynamics into into the game at the same time of course because I haven't yet acted because I've only got prior beliefs in fighting in a form stray beliefs about those about those policies they have to be counterfactual so you cannot get away from this if you want to talk about a perceptual machine that's in a world there's acting upon that world too worth of the data upon which it's making its inferences and doing it's perception you have to have reactionary dispositions you have to have counterfactuals and they have to have some type richness and some depth and I think that's absolutely crucial you can't you know this is something that the formalism I think brings to the tables and now you have to talk about what implications does that have for a sense of self and a sense of agency that is implicit in in the action now these predispositions what should I assume for my prior beliefs about the way that I'm going to act well we already know from there is isn't there and I think it's because it's not connected what are you into oh oh I see sorry didn't right we didn't need to be connected anyway it's a wireless one March so when did I go silence then a long time ago does it matter where were we you know what was I talking about counterfactuals yeah we have to have counterfactuals yes so the the predilections the predispositions to the way that I will act and now yes that's right so the answer is rather obvious because you just go back to Helmholtz or Cantore right through to Geoffrey Hinton if I am doing my job properly then I am optimizing my free energy therefore the only belief that can possibly entertain is that I will behave to maximise my expected free energy it's as simple as that and that's basically what this this expression here is saying it's saying that my beliefs about what I'm going to do my policies my pedis positions my actions are a softmax function of the expected free energy that will accrue under a particular policy in the future notice also this one this is why I knew that I wouldn't finish because there's lots of lovely details there but I just wanted to point out in relation to the argument about lots of models or one big model yeah so I think you're absolutely right to say well it's probably all a question of the structure of the one big model and if you carve the one big model in the right sort of way then they it will look as if there are lots of little models and that's absolutely true in terms of say hierarchical cleaving of the model there's an all another sort of carving into lots of little models which is somebody actually mentioned it in one of the questions about contacts arising or conditioning so notice what I've done here just for simplicity really but it also may have implications for the efficiency of the variational message-passing I'm conditioning states of the world latent or hidden states of the world on a particular all all allowable policies on all these P dispositions that I might entertain because I am the sort of thing that I am so that's and so in a sense you could say each one of these there's one of these belief structures that is associated with every policy that I could entertain so in that sense there are lots and lots of little models and then we're going to do Bayesian model averaging or Bayesian model selection to select the the the correct model and I think that would be your argument in terms that's why there's only one thing in mind at any one particular time in terms of what we're actually doing or sampling or perceiving actively so yeah that Buddhism so this this what I'm going to do now is just focus on this quantity here and this is where the dynamics gets in so you were pressed about how far have people got with variation altering quarters in terms of putting hidden Markov models back in in turn and the question was not very far at the moment so this is I think the next step in terms of putting this variational technology into things like deep learning it's not easy but you have to now entertain objective functions that run into the future and that brings us to again highlights the notion of counterfactuals these are counterfactuals that yet to have happened what we do is appeal to a Hamilton's principle of least action and we sort of take a path integral perspective on this but very very simply what we're going to do is consider beliefs or policies more likely if they maximize a path integral into the future over the expected free energy now remember before the N at the energy was basically accuracy and complexity when we take the expectation under observations that have yet to be secured these two things take very different roles but that you can still see the salaah the homologs in terms of the instantaneous formulation basically our complexity becomes an epistemic value or information gained and the evidence becomes an extrinsic value in in the sense that it's scores the degree to which the expected outcomes will maximize my prior preferences so this is how you get desires preferences goals into the expression they are part of a generic functional that can be cut in terms of extrinsic and intrinsic value or epistemic value or information gained here so this is this is the pragmatic part of it now this is that complexity term where we're now trying to maximize the divergence because the in the past outcomes informed beliefs about hidden states of the world we're in the future beliefs about states of the world will determine outcomes so now what we have is this interesting structure here which is the epistemic value for those who are mathematicians this is just basically the difference between beliefs about hidden or latent states in the world with and without observations and those beliefs can only become more precise more confident with information so it scores the degree to which getting those data those outcomes those observations will reduce my uncertainty about states of the world and that's why it's called a we call it expensive values also called intrinsic motivation in robotics value of information in economics it has a number of different names I just want to quickly go through the special cases of this because if you haven't seen this formalism before it may look a little bit magical but it's not there will be special cases which you're at least all heard of or may actually even recognize formally if I just go through it let me just take this intrinsic value or epistemic value or information gain here I just reproduced it and I've written it out as an expectation of a divergence between a posterior and a prior now if people are familiar with visual search from formal descriptions of visual search and what parts of the sensorium will will be salient enough to specify the focus of the next Koenig eye movement what you'll find is a literature on salience maps that are usually cast in terms of this quantity Bayesian surprised so that makes all of a sense so what we're saying is we want to we can now understand the gathering of those sparse data that maximize my information gain or minimize my uncertainty and that is a deterministic quantity that can be evaluated in terms of this Bayes and surprises by people like eighty and Christoph Kok to describe saccadic searches of our environment this expectation is actually just the mutual information between hidden states of the world and the Associated consequences or causes or outcome so it's just a mutual information between the causes and the consequences under a particular policy so that's nice because that fits in very comfortably with people people like Horace Barlow's assertion that everything we do is in the service of maximizing the mutual information between what's out there and how we record it principle of minimum redundancy or the principle of maximum efficiency them for max principle by Ralph olinska and so on and so on let me put now these prior preferences back in the game but take out a sort of uncertainty the sort of uncertainty which is associated with the fact that I can only see the world in a partial or a sparse way let's assume let's make things simple and assume I can actually see the latent state of the world so there are no hidden States my observations are the states of the world and vice versa what happens them well we just have these two terms but the O's become s's and now we have a very simple expression which we want to minimize hence the minus sign which is the Diver virgins between what we think will happen given this particular policy and what we prefer to happen so the posterior under a policy over outcomes and my preferred outcomes summed over time so this in economics is called a risk sensitive control in optimal control theory it's called KL control because this is a KL kullbackleibler divergence and it's what people would use in lieu R instead of reinforcement learning when they want to put the risk into or accommodate risk so it's used in plant control quite a lot thank you that's well actually I guess it Mike so this was the important stuff from from the point of view of release from from the physicists point of view the rest is just flour is this entertainment apart from this bit so I'm now going to take the last sort of uncertainty out of out of the game and that's basically the relative uncertainty about what would happen if I did that and that brings us now back to just the prior preferences as though in economics that would be expected utility theory and in psychology it might be things like reinforcement learning so you've seen that what I would have shown you is an intuitive example of this intrinsic value the resolution uncertainty you know the epistemic imperatives that drive that sort of good behavior reviews a traffic light example the point basically be you don't need to know the truth state of the world to know exactly how much uncertainty you can resolve if you did that as opposed to this and choosing one or the other is a unitary thing you can only do one thing at a time so I'm surprised you didn't say that you're probably sitting there not thinking about feeling well enough to say this or should I say it anyway that would be and that would that would that would have been something nice to talk about so we've seen this is our hidden Markov model but crucially we have to think about the transitions from latent States h's in martyr slides heir states here that are conditioned upon stuff we can control so these bees now are functions of our policies the transition matrices are functions of our policies that's how we get behavior and action into the game you can write down the genitive bottle for these things in a generic way you can then use off essentially off-the-shelf variational message-passing and then we get to one just considerations about the form of your encoding the processes underlying belief updating how we can understand those in terms of a crude functional anatomy I'm giving you a little example of a mazes little rat in a maze is it initially a very curious rat and goes around working out where through walls are and then it gets bored once it's exhausted all the novelty and environment and then starts become pragmatic and then I've shown you simulations that a neurobiologist would have appreciated play cell activity mismatch negativity is using that simulation and then I would have shown off by showing how you can compose hierarchical Markov models to simulate reading of words by contextualizing them by putting empirical priors over beliefs about narratives or sentences and then showing you some simulations in saccadic eye movements where in we don't look every word we just look at those good words that resolve our uncertainty about the options in our head that best explain what we will see in the future and then I would have taken you through some of the electro physics well this is a belief updating from the simulation emphasizing the separation of temple timescales in terms of these counterfactuals at different levels of deep temporal models and then discuss some of the empirical evidence for that in their notions in silico experiments looking at local and global deviations and compared those to the to the electrophysiological data then ira thanked everybody who has contributed to this talk in terms of their ideas and I would have ended up thanking you for your attention thank you very much [Applause]